

QUESTION 1

Determine whether each of the programming statement(s) in Q1(a)-Q1(j) is VALID or INVALID based on the information given in Figure Q1.1.

```
#include <stdio.h>

int main()
{
    struct Data
    {
        int a, b;
        double c, d;
        char j[20], k;
    };

    struct MyData
    {
        struct Data data;
        struct MyData *p1;
    };

    return 0;
}
```

Figure Q1.1

(10 marks)

No	Programming Statement(s)	Answer
(a)	struct Data data1 = {101, 204, 6.5, 8.5, "Azah", "M"};	
(b)	struct Data data2 = {0, 0, 99.5, 98.5, "Kamal", 'S'};	
(c)	struct Data *myData1, *myData2; struct Data data1, data2; myData1 = &data2; myData2 = myData1;	
(d)	struct Data *myData1, *myData2; struct Data data1, data2; struct Data *myData1, *myData2; myData2 = NULL;	
(e)	struct Data *myData1, *myData2; struct Data data1; struct Data data2 = {0, 0, -0.8, -0.5, "Azza", 'Q'}; data1 = data2; struct Data *myData1, *myData2; struct MyData myData2 = {data1, NULL};	
(f)	struct Data *myData1, *myData2; struct Data data2; struct Data data1 = {0, 0, -99.5, -98.5, "Azza", 'Q'}; data2 = data1; struct MyData myData3 = {data2, NULL};	
(g)	struct Data data2 = {0, 0, 99.5, 98.5, "Kamal", 'S'}; struct MyData myData3 = {data2, NULL}; struct MyData myData3.data.k = "W";	
(h)	struct Data data2 = {0, 0, 99.5, 98.5, "Kamal", 'S'}; struct MyData myData3 = {data2, NULL}; printf("\n%d %.2lf %c", myData3.data.b, myData3.data.d, myData3.data.k);	
(i)	struct Data data2 = {0, 0, 99.5, 98.5, "Kamal", 'S'}; struct MyData myData3 = {data2, NULL}; scanf("%lf %lf %s", myData3.data.c, myData3.data.d, myData3.data.j);	
(j)	struct Data data2 = {0, 0, 99.5, 98.5, "Kamal", 'S'}; struct MyData myData3 = {data2, NULL}; struct MyData *myData4 = myData3;	

QUESTION 2

Determine the output for the program in Figure Q2.1.

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int rewardArray[]={10, 90, 20, 80, 60, 50, 30};

    struct NumberNode{
        int rewardPoint;
        struct NumberNode *next;
    };

    struct NumberNode *front=NULL, *ptr;

    int idx=6;
    while (idx>=0)
    {
        ptr= malloc(sizeof(struct NumberNode));
        if (rewardArray[idx]<=50)
            ptr->rewardPoint = rewardArray[idx]+100;
        else
            ptr->rewardPoint = rewardArray[idx]+1000;

        ptr->next = NULL;

        if (front==NULL)
            front = ptr;
        else
        {
            ptr->next=front;
            front=ptr;
        }

        idx--;
    }

    ptr=front;
    while (ptr!=NULL)
    {
        printf("%5d\t",ptr->rewardPoint);
        ptr=ptr->next;
    }

    return 0;
}
```

Figure Q2.1

(15 marks)

Answer:

QUESTION 3

Azza Tuition Centre has decided to keep its students' records which contain the following data: name, ID number, total number of courses and tuition fee. The declaration of the data to be kept is presented in Figure Q3.1. Answer Q3(a)-Q3(d) based on the information given in Figure Q3.1.

```
#include <stdio.h>
#define SIZE 30
int main()
{
    struct Student{
        char name[20];
        int IDNo;
        int totalCourses;
        double totalFee;
    };

    return 0;
}
```

(a) Declare an array to store records of 30 students for Azza Tuition Centre.

(1 mark)

Answer:

(b) Write a code segment to allow user to enter each student's name, id number and total number of courses enrolled into the array declared for Q3(a).

(6 marks)

Answer:

- (c) Write a code segment to calculate the total tuition fee for each student. The total tuition fee is to be kept in the array. Note that the cost for each course enrolled is RM45.00.

(5 marks)

Answer:

- (d) Write a code segment that count and display students' names and id number with minimum of three courses. Display the total count of the students who enrolled in three courses or more.

(12 marks)

Answer:

- (e) Write a code segment to determine the average total fee for the students at Azza Tuition Centre.

(11 marks)

Answer: